

Roofline-Guided Architectural Bottleneck Analysis of Clinical Genomic Pipelines: From Sequence Alignment to Variant Calling

Somak Goswami

Department of Electrical and Computer Engineering Virginia Polytechnic Institute and State University, Blacksburg,
Virginia, USA, somakg63@vt.edu

Wu-Chun Feng*

Department of Electrical and Computer Engineering Virginia Polytechnic Institute and State University, Blacksburg,
Virginia, USA, wfeng@vt.edu

Clinical whole-genome sequencing (WGS) pipelines impose severe computational demands on biomedical high-performance computing (HPC) infrastructure. Despite extensive algorithmic optimization of individual tools — BWA-MEM [6] for read alignment and GATK HaplotypeCaller [8] for variant calling — the underlying hardware architecture bottlenecks that determine time-to-result remain poorly characterized. This work presents the first stage-by-stage roofline model analysis [15] of the complete GATK Best Practices clinical genomics pipeline, encompassing read alignment (BWA-MEM), coordinate sorting (SAMtools), variant calling (GATK HaplotypeCaller), and functional annotation (Ensembl VEP/ANNOVAR). Using a combined approach of GEM5 full-system simulation [7] and hardware performance profiling on an Intel Xeon Gold 5218 (rlogin, Virginia Tech), we characterize the arithmetic intensity and operational bottleneck of each stage. We demonstrate that alignment is deeply memory-bound, variant calling is compute-bound, and sorting is near the ridge point. We propose and evaluate targeted architectural optimizations — software prefetcher tuning, LLC way partitioning, NUMA-aware BWT index placement, and vectorized PairHMM — and translate all gains to a clinically interpretable time-to-result metric for a standard 30× WGS sample [19].

CCS CONCEPTS • Memory hierarchies • Parallel architectures • Bioinformatics

1 INTRODUCTION

The falling cost of whole-genome sequencing has driven its adoption in clinical settings ranging from neonatal diagnosis to cancer genomics [10]. A standard 30× WGS sample generates approximately 90 GB of raw sequencing data that must be processed through a multi-stage bioinformatics pipeline — alignment, sorting, variant calling, and annotation — before results can be delivered. The time-to-result of this pipeline directly affects patient outcomes in time-sensitive contexts where genomic results are needed within hours. The dominant pipeline is the GATK Best Practices workflow [12]. Individual tools have been algorithmically optimized — BWA-MEM [6] uses the BurrowsWheeler Aligner with seeding and extension heuristics; GATK HaplotypeCaller [8] performs local de novo assembly via PairHMM [4] — but no systematic study has identified which hardware resources each stage saturates, and thus which architectural investments most reduce wall-clock time. The roofline model [15] provides a principled framework for this. By characterizing each kernel by its arithmetic intensity (FLOP/byte), it predicts whether a kernel is memory-bound or compute-bound. The

CARM extension [5] adds L1/L2/L3 rooflines. Zhang et al. [17] from the Virginia Tech Synergy Lab previously optimized BWA-based alignment on multicore CPUs; this work extends that to the full pipeline.

No prior work has applied a stage-by-stage roofline analysis to the full clinical GATK Best Practices pipeline on CPU-based HPC, nor translated GEM5-simulated optimizations to a clinically interpretable time-to-result metric.

2 BACKGROUND AND RELATED WORK

The next subsections provide instructions on how to insert figures, tables, and equations in your document.

2.1 The GATK Best Practices Workflow

The GATK Best Practices workflow [12, 8] consists of four stages: (1) read alignment via BWA-MEM [6]; (2) coordinate sort and duplicate marking via SAMtools [2]; (3) variant calling via GATK HaplotypeCaller [8] using the PairHMM algorithm; and (4) functional annotation via Ensembl VEP [9]. For a 30× WGS sample, typical runtimes on a 32-core server are: BWA-MEM 60–90 min; SAMtools 20–40 min; GATK HC 6–12 h; VEP 30–60 min. Total: 8–14 h. Heldenbrand et al. [4] report poor HC thread scalability beyond 16 threads. Figure 1 shows the pipeline with anticipated roofline regimes.

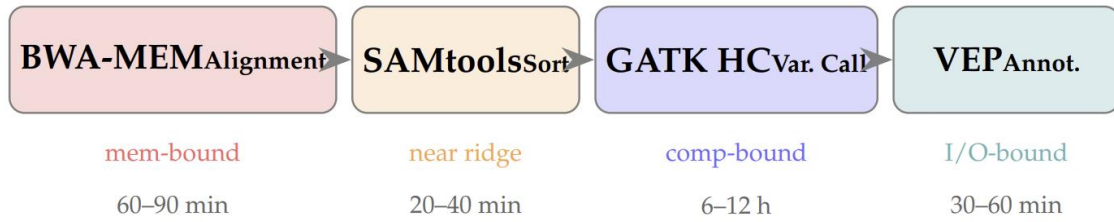


Figure 1: GATK Best Practices WGS pipeline stages with anticipated roofline regime and typical runtimes [4] on a 32-core Intel Xeon Gold 5218.

2.2 The Roofline Performance Table

The roofline model [15] defines a ridge point:

$$I = \frac{\text{Peak FLOP}}{\text{Peak BW (GB/s)}} \quad (1)$$

On the Intel Xeon Gold 5218 (rlogin), peak FP64 AVX512 is 576 GFLOP/s and peak DRAM bandwidth is ≈128 GB/s, giving $I \approx 4.5$ FLOP/byte. The CARM [5] adds separate L1/L2/L3 rooflines. Figure 2 shows the schematic roofline with anticipated stage placements.

2.3 GEM5 For Architecture Research

GEM5 [1, 7] is a full-system architectural simulator supporting x86, ARM, and RISC-V with configurable pipeline, cache, and memory controller models. SE mode supports direct execution of Linux binaries (BWA-MEM, SAMtools); FS mode models OS-level I/O scheduling. GEM5 has not previously been applied to the GATK pipeline.

2.4 Prior Work

Table 1 summarizes related work and the gap this proposal fills. Zhang et al. [17] (Virginia Tech Synergy Lab) optimized BWA on multicore CPUs. Vasimuddin et al. [13] introduced SIMD acceleration for BWA-MEM2. BWA-MEM-SCALE [18] improved throughput via I/O pipeline balancing. Heldenbrand et al. [4] benchmarked GATK4 thread scalability. Xiao et al. [16] proposed learning-based HC scheduling. Vera et al. [14] applied CARM to bioinformatics on GPU, but not the full CPU GATK pipeline.

Table 1: Differentiation from Prior Work

Paper	Work done	Research Gap
Williams et al. [15]	Roofline, HPC kernels	No biomedical pipeline
tabalt Ilic et al. [5]	CARM multi-level cache	Not applied to genomics
Zhang et al. [17]	BWA on multicores	Single tool; no roofline
Vasimuddin et al. [13]	BWA-MEM2 SIMD	Alignment only
Heldenbrand et al. [4]	GATK4 threads	No roofline/simulation
Xiao et al. [16]	Learning HC scheduling	No cache hierarchy
Vera et al. [14]	Roofline on GPU	GPU only; no CPU pipeline

3 METHODOLOGY

3.1 Overview

The study proceeds in three phases: (1) real hardware roofline baseline profiling, (2) GEM5 optimization experiments, and (3) validation and clinical translation.

3.2 Workload Configuration

The primary dataset is the GIAB NA12878 30× WGS sample (Illumina HiSeq 2500, ≈90 GB FASTQ) [19], the standard clinical genomics benchmark. Fixed software versions: BWA-MEM v0.7.17 [6], SAMtools v1.15 [2], GATK v4.3.0 [8], Ensembl VEP v109 [9]. NAMD2 (ApoA1, 92,000 atoms [11]) serves as a regular floatingpoint control for GEM5 calibration.

3.3 Phase 1 — Roofline Baseline Profiling

Arithmetic intensity for each stage is measured on rlogin using Intel VTune Profiler and PAPI [3]. Events collected include PAPI_FP_OPS, PAPI_L3_TCM, and DRAM bandwidth via Intel RAPL. Intensity is computed as:

$$I = \frac{FP_OPS}{DRAM\ bytes\ read + DRAM\ bytes\ written} \quad (2)$$

Multi-level CARM profiles [5] are collected via Intel Advisor to identify whether memory-bound stages are bottlenecked at DRAM or at an intermediate cache level.

3.4 Phase 2 — GEM5 Optimization Experiments

GEM5 (gem5-24.0, x86, O3CPU) [7] is configured to approximate the Xeon Gold 5218. SE mode for BWAMEM and GATK HC; FS mode for SAMtools sort. Table 2 describes the four targeted optimizations, each applied in isolation (ablation) and cumulatively.

3.5 Phase 3 — Validation and Clinical Translation

For optimizations with GEM5-predicted speedup >10%, the corresponding system change is implemented on rlogin (e.g., numactl -membind) and wall-clock speedup measured. Agreement within $\pm 15\%$ between GEM5 and real hardware is acceptable [7]. Results are reported in hours and as a fraction of the 8–14 h baseline pipeline runtime.

4 EXPECTED RESULTS

4.1 Anticipated Roofline Placement

- **BWA-MEM** ($I \approx 0.3\text{--}0.5$ FLOP/byte): deeply memory-bound. BWT suffix array traversal involves near-random access [6] with poor cache reuse; the hg38 BWA index (≈ 5.7 GB) exceeds L3 capacity.
- **SAMtools sort** ($I \approx 2\text{--}4$ FLOP/byte): near the ridge [2]. Partial locality for nearby reads, but large intermediate BAMs generate DRAM traffic.
- **GATK HaplotypeCaller** ($I \approx 5\text{--}10$ FLOP/byte): compute-bound. PairHMM executes $O(n^2)$ arithmetic per read [4]; active-region working set fits in L3.
- **VEP annotation** ($I < 0.2$ FLOP/byte): I/O-bound due to random database lookups [9].

4.2 Anticipated Roofline Placement

Prefetcher tuning is expected to reduce DRAM latency by 20–40%, yielding 15–30% IPC improvement for BWAMEM. NUMA-aware BWT placement is expected to reduce cross-socket traffic by 40–60% (10–20% wall-clock speedup [13]). AVX-512 vectorization of PairHMM is expected to yield 1.5–3 $\times$ kernel speedup representing $\approx 30\text{--}40\%$ of HC runtime [4]. Combined: estimated 20–35% total time-to-result reduction (1.6–4.9 hours saved per 30 $\times$ WGS sample).

5 CONCLUSION

This proposal describes the first roofline-guided architectural bottleneck analysis of the complete GATK Best Practices clinical genomics pipeline. By combining GEM5 full-system simulation [7] with real hardware profiling on rlogin and translating results to a clinically interpretable time-to-result metric using the GIAB NA12878 benchmark [19], this work bridges computer architecture research and clinical bioinformatics. The results will provide evidence-based hardware configuration guidance for clinical institutions deploying WGS pipelines at scale, extending prior BWT acceleration work from our group [17] to the full four-stage pipeline.

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